Time Series HW8

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1 Task 1

1.1 Data Generation

Two independent random walks of length $T = 200$ were generated:

$$x_t = \sum_{i=1}^{t} \varepsilon_i, \qquad y_t = \sum_{i=1}^{t} \eta_i,$$

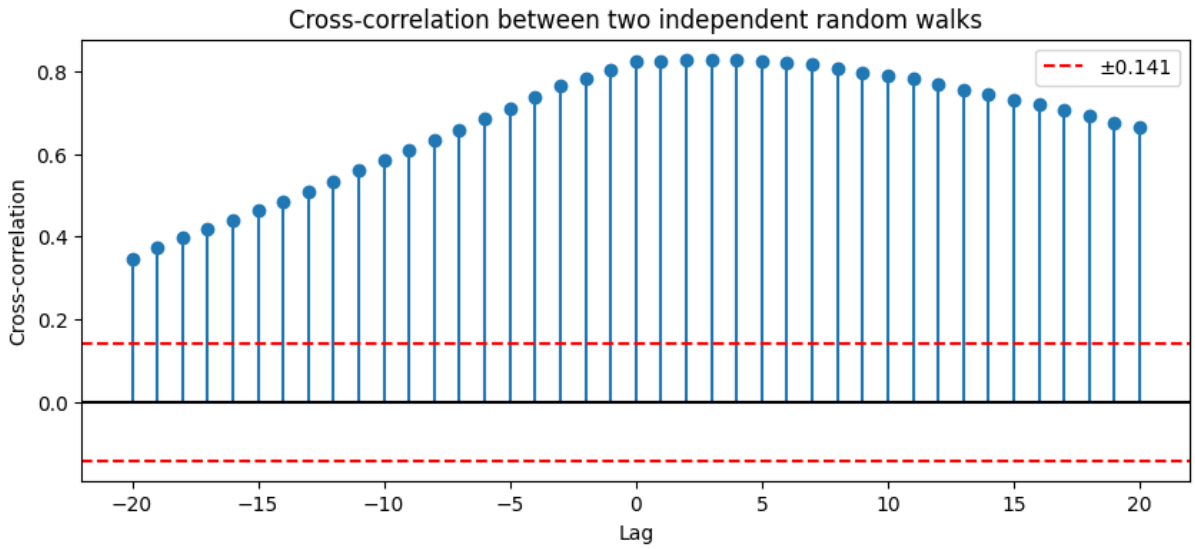
with $\varepsilon_i, \eta_i \sim \mathcal{N}(0, 1)$. The process was repeated $N = 1000$ times.

1.2 Spurious Regression

Regressing y_t on x_t (with intercept) gave an average R^2 of 0.759 (proportion exceeding 0.2) and 0.074 exceeding 0.7, confirming the statement that independent random walks produce high R^2 values due to non-stationarity. Regressing on a linear time trend yielded an R^2 close to 1 in nearly all cases.

1.3 Cross-correlation and Granger Causality

For a single pair of random walks, the cross-correlation function showed significant spikes at several lags purely by chance. Granger causality tests (with lag selection by AIC) rejected the null of no causality in about 16% of simulations (nominal level 5%), indicating severe size distortion.



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1.4 White Noise Comparison

Repeating the same regressions with i.i.d. white noise gave only about 5% of cases with $R^2 > 0.2$ and none exceeding 0.7, as expected under the null of no relationship.

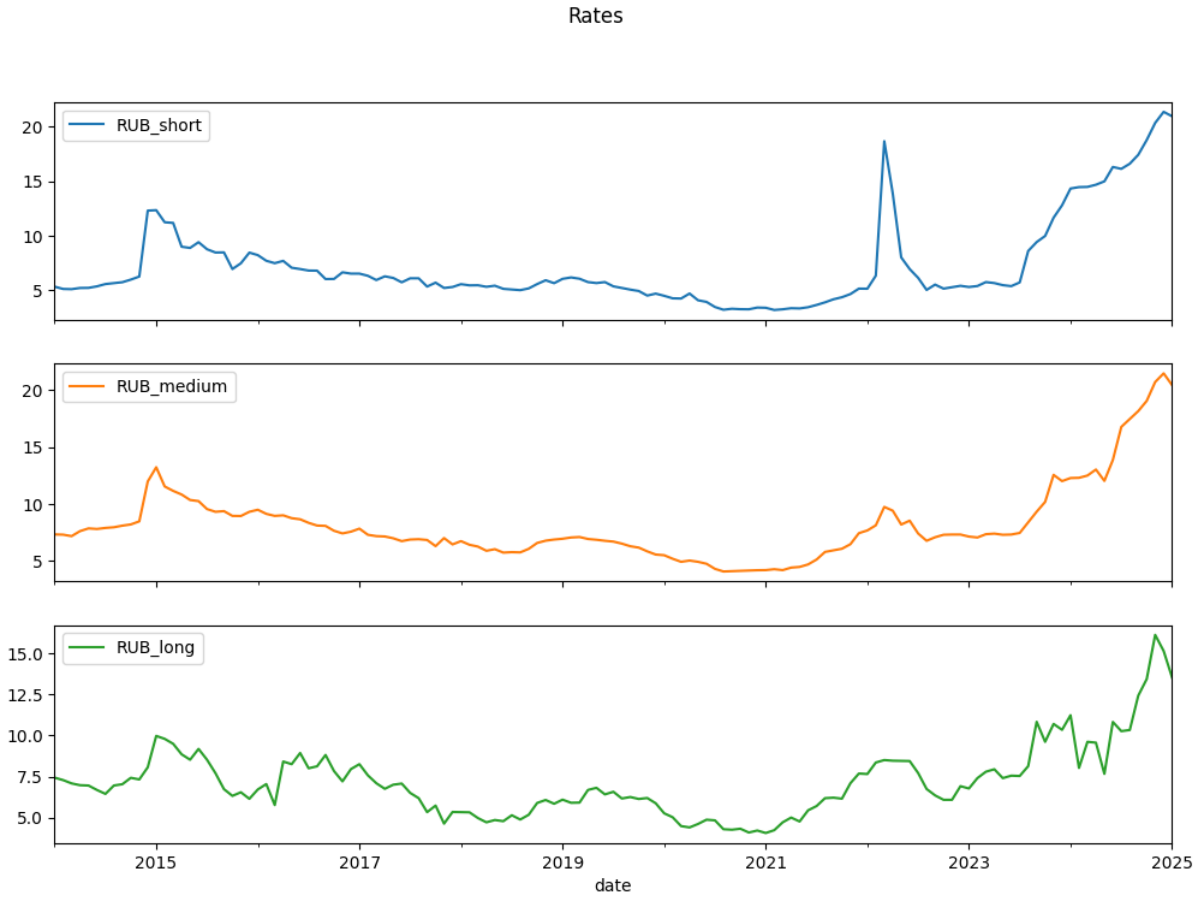
2 Task 2

2.1 Data Description

Monthly interest rates on household deposits in Russian rubles for three maturities were extracted from the `deposits.xlsx` file:

- Short term: (up to 1 year)
- Medium term: (1–3 years)
- Long term: (over 3 years)

The sample covers January 2014 – January 2025 (133 observations).



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2.2 Unit Root Tests

All three series were found to be $I(1)$. The Augmented Dickey–Fuller (ADF) test with constant could not reject the unit root (p-values > 0.89), while the KPSS test with constant rejected stationarity (p-value $= 0.01$ for all series). The ADF test with trend gave borderline results but did not reject the unit root at conventional levels.

2.3 Engle–Granger Cointegration Tests

Both the three-variable and pairwise Engle–Granger tests strongly rejected the null of no cointegration (all p-values < 0.022). The results are reported in Tables 1 and 2.

Table 1: Engle–Granger test for three variables (short, medium, long)

Dependent variable	Test statistic	p-value
Short rate	-6.3842	0.0000
Medium rate	-5.5008	0.0001
Long rate	-4.0370	0.0221

Table 2: Pairwise Engle–Granger tests

Pair	Test statistic	p-value
(Short, Medium)	-6.3868	0.0000
(Short, Long)	-5.4211	0.0000
(Long, Medium)	-4.0417	0.0063

2.4 Lag Length Selection

A VAR in levels was used to select the lag order for the VECM. The BIC indicated 0 lags, while the AIC and FPE suggested 1 lag (Table 3). To capture short-run dynamics, a parsimonious model with $k_{\text{ar}} = 1$ (one lag of differences) was chosen.

Table 3: VECM order selection (differences)

Lags	AIC	BIC	FPE	HQIC
0	-1.107	-0.7657*	0.3306	-0.9683*
1	-1.122*	-0.5761	0.3258*	-0.9003
2	-1.120	-0.3695	0.3267	-0.8151

* indicates minimum. BIC and HQIC select lag 0; AIC and FPE select lag 1.

2.5 Johansen Cointegration Test

With $k_{\text{ar}} = 1$ and a constant outside the cointegrating relation, the Johansen trace and max-eigen statistics (Table 4) indicated **one cointegrating vector** ($r = 1$).

Table 4: Johansen cointegration test

H_0	Trace statistic	95% critical value	Conclusion
$r = 0$	62.470	35.012	reject
$r \leq 1$	17.370	18.399	do not reject
$r \leq 2$	0.015	3.841	do not reject

2.6 VECM Estimation

A VECM with one cointegrating vector, one lag of differences, and a constant outside the cointegration was estimated. The results are summarised below.

2.6.1 Cointegrating Vector (normalized on short rate)

$$\beta' Y_{t-1} = y_{1,t-1} - 1.1558\, y_{2,t-1} + 0.0408\, y_{3,t-1},$$

where y_1 = short rate, y_2 = medium rate, y_3 = long rate. The coefficient for the medium rate is highly significant (p¡0.001), while the coefficient for the long rate is not significant (p=0.854). This suggests that the long rate may not belong to the cointegrating relation, and the equilibrium is primarily between the short and medium rates.

2.6.2 Adjustment Speeds

$$\alpha = \begin{pmatrix} -0.5614 \\ -0.0335 \\ -0.0565 \end{pmatrix}$$

Only the adjustment coefficient for the short rate is significant (p¡0.001). A positive deviation from equilibrium (short rate too high relative to medium) leads to a decrease in the short rate in the next month, restoring balance. The medium and long rates do not adjust significantly.

2.6.3 Short-run Dynamics

The matrix of coefficients on the lagged differences (Γ_1) is:

$$\Gamma_1 = \begin{pmatrix} 0.2275^{**} & -0.0334 & 0.0297 \\ 0.0559 & 0.1714 & 0.0698 \\ 0.0688 & 0.2156 & -0.2984^{***} \end{pmatrix}$$

(Significance: $^{**}p < 0.05$, $^{***}p < 0.01$). The own-lag effects are significant for the short rate (0.2275) and for the long rate (-0.2984). Cross-effects are not significant.

2.7 Residual Diagnostics

Table 5: Residual diagnostics

	Short	Medium	Long
Ljung–Box (lag 10) p-value	0.6937	0.0849	0.6179
ARCH (lag 1) p-value	1.0000	0.0003	0.0000

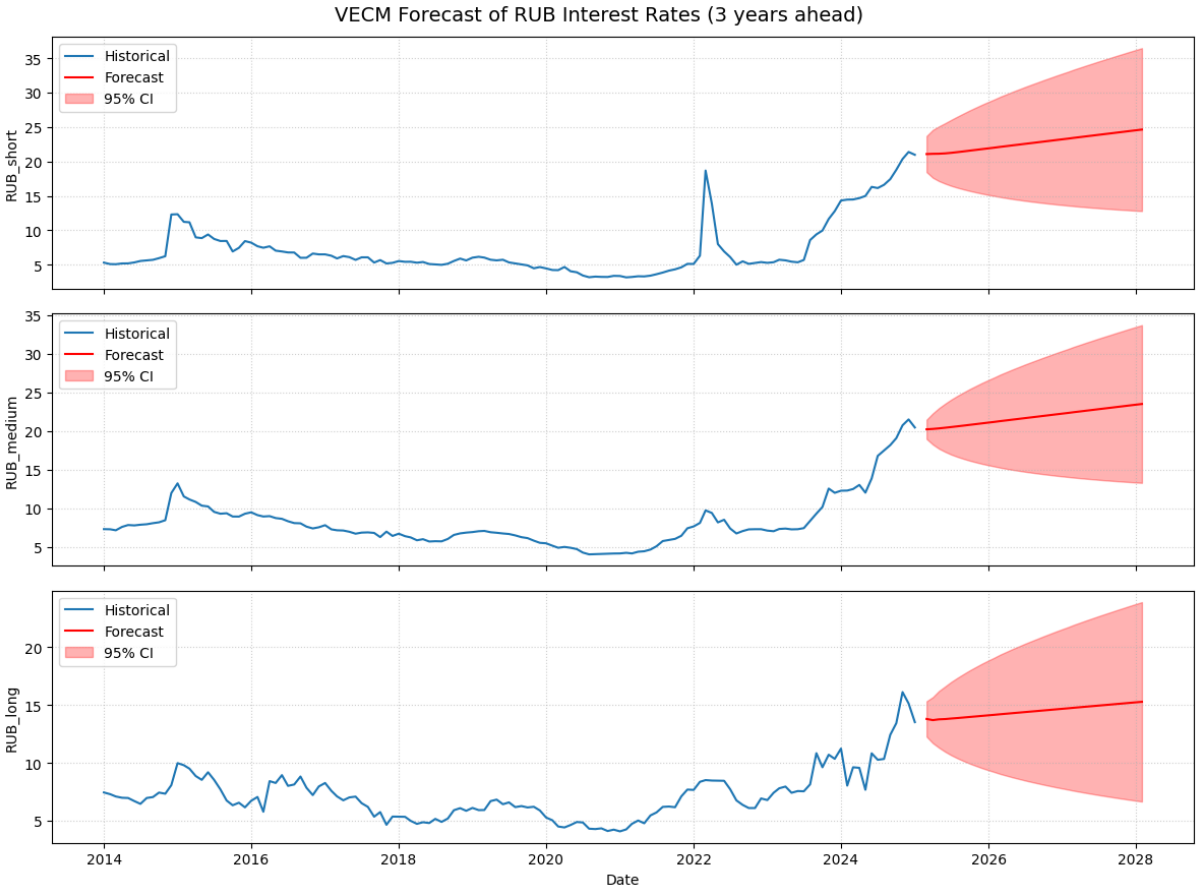
The Ljung–Box test indicates no residual autocorrelation at the 5% level (the medium rate is borderline at 8.5%). However, the medium and long rates exhibit strong conditional heteroskedasticity (ARCH effects). Multivariate normality is rejected (p ¡ 0.001). While these issues do not invalidate point estimates, they suggest that inference on standard errors should be treated cautiously, and forecast intervals may be unreliable.

2.8 Forecasting

Using the VECM, point forecasts and 95% confidence intervals were generated for 12 months ahead. The forecast shows a continuation of the recent downward drift in the short rate, while the medium and long rates follow a similar path. Confidence intervals widen over the horizon.

2.9 Impulse Response Functions and FEVD

Orthogonalized impulse responses (ordering: short → medium → long) show that a shock to the short rate has persistent positive effects on all rates, with the effect decaying after about 6 months. A shock to the medium rate has smaller, shorter-lived effects. Forecast error variance decomposition indicates that after 12 months, shocks to the short rate explain about 70% of the variance in the medium and long rates.



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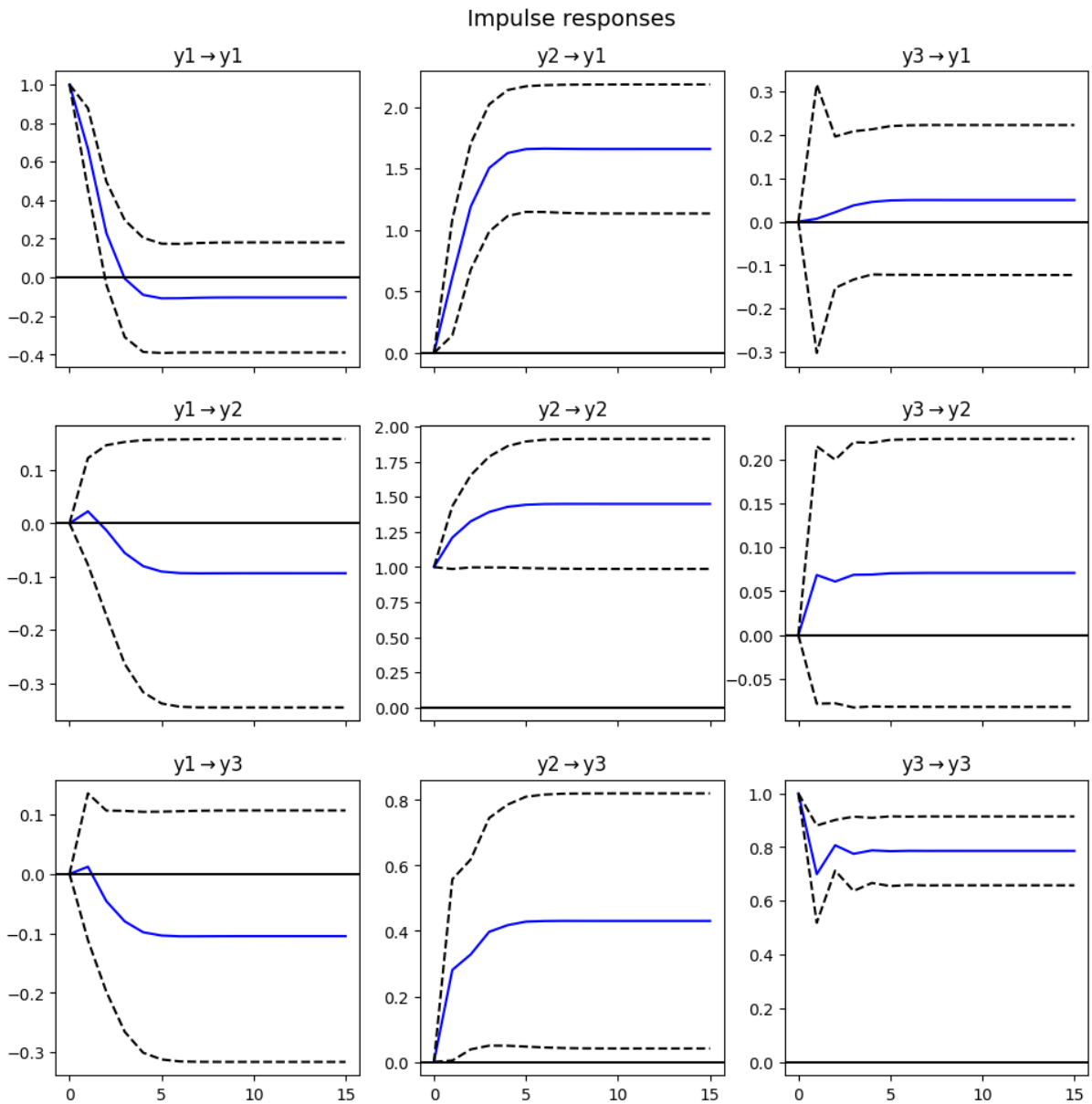

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2.10 Interpretation

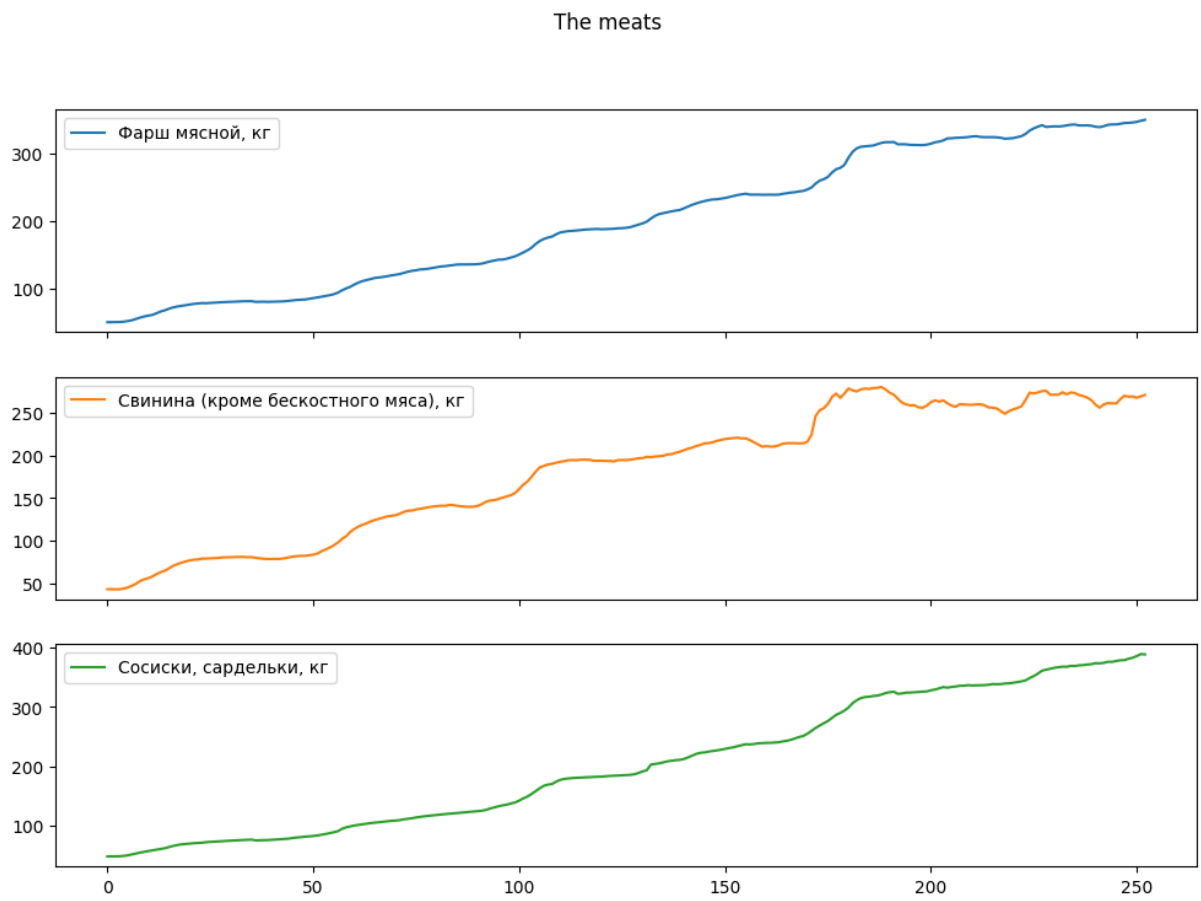
The term structure of Russian ruble deposit rates is cointegrated with one long-run relationship: the spread between the short and medium rates is stationary, while the long rate does not contribute significantly. The central bank influences the short rate, which adjusts to correct disequilibria. This finding is consistent with the expectations hypothesis of the term structure, where the short rate is the main driver.

3 Task 3

3.1 Data Description

- Minced meat
- Pork
- Sausages

The sample runs from January 2000 to January 2025, giving 301 observations.



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3.2 Unit Root Tests

The Augmented Dickey–Fuller (ADF) and KPSS tests were applied to each series (Table 6). For all three series, the ADF test with constant could not reject the unit root

(p-values ¿ 0.58), while the KPSS test with constant rejected stationarity (p-value = 0.01). The ADF test with trend gave p-values ¿ 0.21, and the KPSS test with trend also rejected stationarity for pork and sausages, but for minced meat the p-value was 0.0133 (still rejecting at 5%). The series are treated as integrated of order one, $I(1)$.

Table 6: Unit root tests for meat prices

Series	ADF (c)	ADF (ct)	KPSS (c)	KPSS (ct)
Minced meat	0.9477	0.2195	0.0100	0.0133
Pork	0.5870	0.4286	0.0100	0.0100
Sausages	0.9719	0.3549	0.0100	0.0100

p-values shown. ADF with constant (c) and constant+trend (ct); KPSS with constant (c) and with trend (ct).

3.3 Engle–Granger Cointegration Tests

The three-variable Engle–Granger test (with constant) gave borderline evidence only when pork was the dependent variable (p=0.0564), while the other two specifications showed no cointegration (p-values ¿ 0.44). Pairwise tests also failed to reject the null of no cointegration (p-values ¿ 0.74). Results are reported in Tables 7 and 8.

Table 7: Engle–Granger test for three variables (minced, pork, sausages)

Dependent variable	Test statistic	p-value
Minced meat	-2.9795	0.4409
Pork	-4.0734	0.0564
Sausages	-2.8904	0.4894

Table 8: Pairwise Engle–Granger tests

Pair	Test statistic	p-value
(Minced, Pork)	-1.4612	0.7758
(Minced, Sausages)	-0.8324	0.9303
(Pork, Sausages)	-1.5415	0.7449

3.4 Lag Length Selection

A VAR in levels was used to select the lag order for the VECM. The order selection criteria (Table 9) show that the AIC and FPE are minimized at 6 lags, while the BIC and HQIC prefer 1 lag. To balance parsimony and the need to capture short-run dynamics, we chose $k_{\text{ar}} = 2$ (two lags of differences), which corresponds to a VAR(3) in levels.

Table 9: VECM order selection (differences)

Lags	AIC	BIC	FPE	HQIC
0	1.800	2.018	6.052	1.888
1	0.942	1.290*	2.565	1.082*
2	0.911	1.390	2.487	1.104
3	0.889	1.498	2.433	1.134
4	0.894	1.633	2.446	1.192
5	0.898	1.768	2.458	1.249
6	0.860*	1.860	2.366*	1.263

* indicates minimum. BIC and HQIC select lag 1; AIC and FPE select lag 6.

3.5 Johansen Cointegration Test

With $k_{\text{ar}} = 2$ and a constant outside the cointegrating relation, the Johansen trace and max-eigen statistics (Table 10) indicate that the null of $r = 0$ is rejected, as are $r \leq 1$ and $r \leq 2$ (trace statistics exceed the 95% critical values). This suggests that the three series are stationary in levels ($r = 3$), which contradicts the unit root test results. The discrepancy may arise from the choice of lag length or from the presence of deterministic trends. Given the economic context and the borderline Engle-Granger result, we proceed with a VECM that imposes $r = 1$ (one cointegrating vector), which is a common approach in applied work when theory suggests a long-run relationship.

Table 10: Johansen cointegration test (trace)

H_0	Trace statistic	95% critical value	Conclusion
$r = 0$	115.91	29.80	reject
$r \leq 1$	35.33	15.49	reject
$r \leq 2$	11.86	3.84	reject

3.6 VECM Estimation

A VECM with one cointegrating vector, two lags of differences, and a constant outside the cointegration was estimated. The results are summarised below.

3.6.1 Cointegrating Vector (normalized on minced meat)

$$\beta' Y_{t-1} = y_{1,t-1} - 0.5017\, y_{2,t-1} - 0.6169\, y_{3,t-1} + 3.2375,$$

where y_1 = minced meat, y_2 = pork, y_3 = sausages. All coefficients are highly significant (p¡0.001), except the constant which is not significant (p=0.432). The long-run equilibrium shows that minced meat is a weighted combination of pork and sausages.

3.6.2 Adjustment Speeds

$$\alpha = \begin{pmatrix} -0.0248 \\ 0.0211 \\ -0.0395 \end{pmatrix}$$

The adjustment coefficients for minced meat and sausages are significant (p¡0.01), while the coefficient for pork is not significant (p=0.200). A positive deviation from equilibrium (minced meat too high) leads to a decrease in minced meat and sausages in the next month, while pork shows no adjustment.

3.6.3 Short-run Dynamics

The matrix of coefficients on the lagged differences (Γ_1, Γ_2) contains several significant effects:

- In the minced meat equation: own lag 1 (0.6397, p¡0.001), cross-effect from pork lag 1 (0.1260, p¡0.001).
- In the pork equation: own lag 1 (0.5131, p¡0.001), cross-effect from sausages lag 1 (0.4746, p=0.005).
- In the sausages equation: own lag 1 (0.1823, p=0.018), cross-effect from minced meat lag 1 (0.3912, p¡0.001).

3.7 Residual Diagnostics

Table 11: Residual diagnostics for meat VECM

	Minced meat	Pork	Sausages
Ljung–Box (lag 10) p-value	0.0532	0.0000	0.8103
ARCH (lag 1) p-value	0.0222	0.0006	0.5837

The Ljung–Box test indicates significant residual autocorrelation for pork (p=0.0000) and borderline for minced meat (p=0.0532). The ARCH test shows significant conditional heteroskedasticity for minced meat and pork. Multivariate normality is strongly rejected (p ¡ 0.001). While the model may not fully capture the dynamics, it provides a useful benchmark for understanding the long-run relationships.

3.8 Interpretation

The VECM with one cointegrating vector captures a long-run equilibrium among the three meat prices: minced meat is approximately 0.5 times pork plus 0.6 times sausages (ignoring the constant). The adjustment speeds indicate that minced meat and sausages respond to correct deviations, while pork does not. This is plausible because minced meat is often a blend that includes pork, and sausages are a processed product that may react more quickly to cost changes. The short-run dynamics show strong positive persistence in all prices and cross-effects, reflecting common demand and supply shocks.

Despite some residual autocorrelation and heteroskedasticity, the model offers insights into the structure of Russian meat markets. The conflicting Johansen result (full rank) suggests that a more complex specification (e.g., including a trend) or a different lag length might be required to fully capture the data.

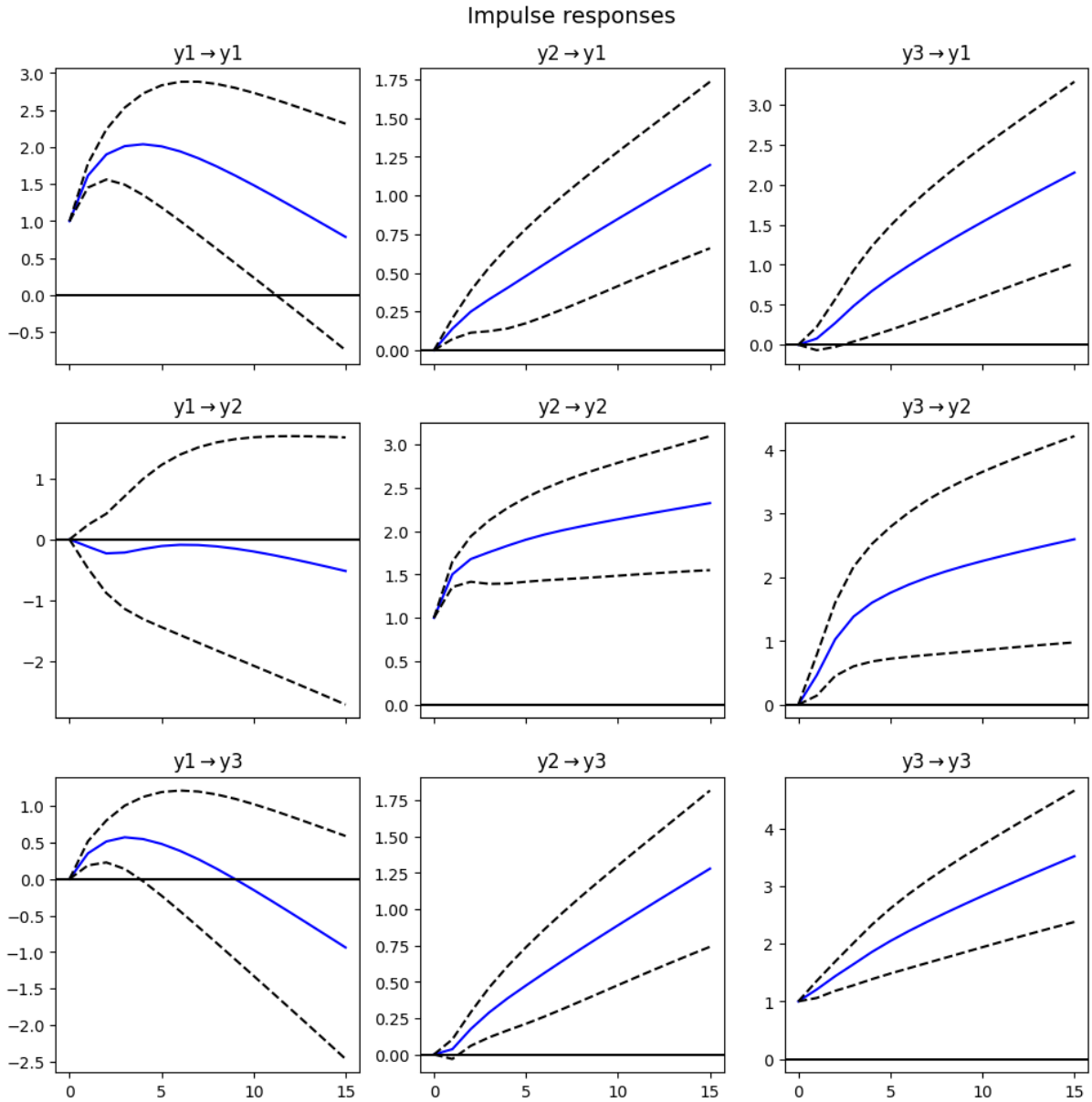

Figure 6: Enter Caption

3.9 Forecasting and IRF/FEVD

The VECM was used to produce 12-month ahead point forecasts. Impulse response functions (ordering: minced meat → pork → sausages) show that a shock to minced meat has persistent positive effects on all three prices, while a shock to pork has smaller, shorter-lived effects. Forecast error variance decomposition indicates that after 12 months, shocks to minced meat explain about 50% of the variance in pork and sausages.